

11. Cotangent bundle and 1-forms

Recall that if V is a finite-dimensional $\mathbb{R}$ -vector space then the space of all linear functionals, i.e., linear maps $\omega : V \rightarrow \mathbb{R}$ is denoted by V^* and is called the dual space to V . Elements of V^* are called **covectors** on V . In fact, we immediately get a basis of V^* once we choose a basis of V as given in the next proposition.

Proposition 11.1. *Let V be a finite-dimensional vector space. Given any basis $(e_1, \dots, e_n)$ for V , let $\varepsilon^1, \dots, \varepsilon^n \in V^*$ be the covectors defined by*

$$\varepsilon^i(e_j) = \delta_j^i,$$

where δ_j^i is the Kronecker delta symbol. Then $(\varepsilon^1, \dots, \varepsilon^n)$ is a basis for V^ , called the **dual basis** to (e_j) and hence $\dim V^* = \dim V$.*

So, for instance, for the standard basis $(e_1, \dots, e_n)$ of $\mathbb{R}^n$, the dual basis is denoted by $(e^1, \dots, e^n)$, and is called the **standard dual basis**. Notice that we are using upper indices for denoting the covectors. These basis covectors are the linear functionals on $\mathbb{R}^n$ given by

$$e^i(v) = e^i(v^1, \dots, v^n) = v^i.$$

In matrix notation, a linear map from $\mathbb{R}^n$ to $\mathbb{R}$ is represented by a $1 \times n$ matrix, that is, it is a **row matrix**. The basis covectors can therefore also be thought of as the linear functionals represented by the row matrices

$$e^1 = (1 \ 0 \ \dots \ 0), \quad e^2 = (0 \ 1 \ 0 \ \dots \ 0), \quad \dots, \quad e^n = (0 \ \dots \ 0 \ 1).$$

In general, if (e_j) is a basis for V and (ε^i) is its dual basis, then for any vector $v = v^j e_j \in V$, we have

$$\varepsilon^i(v) = v^j \varepsilon^i(e_j) = v^j \delta_j^i = v^i.$$

We can express an arbitrary covector $\omega \in V^*$ in terms of the dual basis as

$$\omega = \omega_i \varepsilon^i, \tag{11.1}$$

where the components are determined by $\omega_i = \omega(e_i)$ and similarly, the action of ω on a vector $v = v^j e_j$ is

$$\omega(v) = \omega_i v^i. \tag{11.2}$$

Notation: We always write basis covectors with upper indices, and components of a covector with lower indices.

Suppose V and W are vector spaces and $L : V \rightarrow W$ is a linear map. We define a linear map $L^* : W^* \rightarrow V^*$, called the **dual map** or **transpose of L** , by

$$(L^* \omega)(v) = \omega(Lv) \quad \text{for } \omega \in W^*, v \in V.$$

The dual maps satisfies the following properties.

- (1) $(A \circ B)^* = B^* \circ A^*$.
- (2) $(\text{Id}_V)^* : V^* \rightarrow V^*$ is the identity map of V^* .

For a vector space V , we also have a **second dual space** $V^{**} = (V^*)^*$. For each vector space V there is a natural, basis-independent map $\xi : V \rightarrow V^{**}$, defined as follows. For each vector $v \in V$, define a linear functional $\xi(v) : V^* \rightarrow \mathbb{R}$ by

$$\xi(v)(\omega) = \omega(v) \quad \text{for } \omega \in V^*.$$

Proposition 11.2. *For any finite-dimensional vector space V , the map $\xi : V \rightarrow V^{**}$ is an isomorphism.*

Proof. Because $\dim V = \dim V^{**}$, it suffices to verify that ξ is injective. Suppose $v \in V$ is not zero. Extend v to a basis $(v = e_1, \dots, e_n)$ for V , and let $(\varepsilon^1, \dots, \varepsilon^n)$ denote the dual basis for V^* . Then $\xi(v) \neq 0$ because

$$\xi(v)(\varepsilon^1) = \varepsilon^1(v) = \varepsilon^1(e_1) = 1.$$

□

The preceding proposition shows that when V is finite-dimensional, we can unambiguously identify V^{**} with V itself, because the map ξ is canonically defined, without reference to any basis. It is important to observe that although V^* is also isomorphic to V (for the simple reason that any two finite-dimensional vector spaces of the same dimension are isomorphic), there is no *canonical* isomorphism $V \cong V^*$.

11.1 Tangent Covectors on Manifolds

Now let M be a smooth manifold. For each $p \in M$, we define the **cotangent space at p** , denoted by T_p^*M , to be the dual space to T_pM

$$T_p^*M = (T_pM)^*.$$

Elements of T_p^*M are called **tangent covectors at p** , or just **covectors at p** .

Let (U, x^i) be a coordinate chart. Then the coordinate basis $(\frac{\partial}{\partial x^i}|_p)$ gives rise to a dual basis for T_p^*M , which we denote by $(\lambda^i|_p)$. Any covector $\omega \in T_p^*M$ can thus be written uniquely as $\omega = \omega_i \lambda^i|_p$, where

$$\omega_i = \omega \left(\frac{\partial}{\partial x^i} \Big|_p \right).$$

Let us try to understand how the covectors look like under change of coordinate system. Suppose that $(\tilde{x}^j)$ is another set of smooth coordinates whose domain contains p , and let $(\tilde{\lambda}^j|_p)$ denote the basis for T_p^*M dual to $(\frac{\partial}{\partial \tilde{x}^j}|_p)$. We recall that the coordinate vector fields transform as follows:

$$\frac{\partial}{\partial x^i} \Big|_p = \frac{\partial \tilde{x}^j}{\partial x^i}(p) \frac{\partial}{\partial \tilde{x}^j} \Big|_p. \quad (11.3)$$

Writing ω in both systems as $\omega = \omega_i \lambda^i|_p = \tilde{\omega}_j \tilde{\lambda}^j|_p$, we can use (11.3) to compute the components ω_i in terms of $\tilde{\omega}_j$:

$$\omega_i = \omega \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \omega \left(\frac{\partial \tilde{x}^j}{\partial x^i}(p) \frac{\partial}{\partial \tilde{x}^j} \Big|_p \right) = \frac{\partial \tilde{x}^j}{\partial x^i}(p) \tilde{\omega}_j. \quad (11.4)$$

Thus, the components of a vector, i.e., the n -tuples $(v^1, \dots, v^n)$ and $(\tilde{v}^1, \dots, \tilde{v}^n)$ assigned to two different coordinate systems (x^i) and $(\tilde{x}^j)$ were related by the transformation law

$$\tilde{v}^j = \frac{\partial \tilde{x}^j}{\partial x^i}(p) v^i,$$

and a covector transforms, according to the following slightly different rule:

$$\omega_i = \frac{\partial \tilde{x}^j}{\partial x^i}(p) \tilde{\omega}_j. \quad (11.7)$$

11.2 Covector Fields or 1-forms

For any smooth manifold M the disjoint union

$$T^*M = \coprod_{p \in M} T_p^*M$$

is called the **cotangent bundle** of M . It has a natural projection map $\pi: T^*M \rightarrow M$ sending $\omega \in T_p^*M$ to $p \in M$. As above, given any smooth local coordinates (x^i) on an open subset $U \subseteq M$, for each $p \in U$ we denote the basis for T_p^*M dual to $(\frac{\partial}{\partial x^i}|_p)$ by $(\lambda^i|_p)$. This defines n maps $\lambda^1, \dots, \lambda^n: U \rightarrow T^*M$, called **coordinate covector fields**. Just like the case for tangent bundle Theorem 4.8, the next result states that the cotangent bundle is also a $2n$ -dimensional manifold with a smooth structure inherited from M .

Theorem 11.3. *Let M be a smooth n -manifold. The cotangent bundle T^*M has a unique topology and smooth structure making it into a smooth $2n$ -dimensional manifold.*

Proof. We leave the proof as an exercise. □

As in the case of the tangent bundle, smooth local coordinates for M yield smooth local coordinates for its cotangent bundle. If (x^i) are smooth coordinates on an open subset $U \subseteq M$, then the map from $\pi^{-1}(U)$ to $\mathbb{R}^{2n}$ given by

$$\xi_i \lambda^i|_p \mapsto (x^1(p), \dots, x^n(p), \xi_1, \dots, \xi_n)$$

is a smooth coordinate chart for T^*M . We call (x^i, ξ_i) the **natural coordinates** for T^*M associated with (x^i) .

Again, as was the case with vector fields, we have the following definition.

Definition 11.4. A **covector field** or a **(differential) 1-form** is a continuous map $\omega: M \rightarrow T^*M$ such that

$$\pi \circ \omega = \text{id}_M.$$

Thus, for every $p \in M$, $\omega_p \in T_p^*M$.

In any smooth local coordinates on an open subset $U \subseteq M$, a covector field ω can be written in terms of the coordinate covector fields (λ^i) as $\omega = \omega_i \lambda^i$ for n functions $\omega_i: U \rightarrow \mathbb{R}$ called the **component functions** of ω . They are characterized by

$$\omega_i(p) = \omega_p \left(\frac{\partial}{\partial x^i} \Big|_p \right).$$

If ω is a covector field and X is a vector field on M , then we can form a function $\omega(X): M \rightarrow \mathbb{R}$ by

$$\omega(X)(p) = \omega_p(X_p), \quad p \in M.$$

If we write $\omega = \omega_i \lambda^i$ and $X = X^j \frac{\partial}{\partial x^j}$ in terms of local coordinates, then $\omega(X)$ has the local coordinate representation $\omega(X) = \omega_i X^i$.

Just as in the case of vector fields, there are several ways to check for smoothness of a covector field.

Proposition 11.5. *Let M be a smooth manifold and let $\omega: M \rightarrow T^*M$ be a covector field. The following are equivalent:*

1. ω is smooth.
2. In every smooth coordinate chart, the component functions of ω are smooth.
3. For every smooth vector field $X \in \mathfrak{X}(M)$, the function $\omega(X)$ is smooth on M .
4. For every open subset $U \subseteq M$ and every smooth vector field X on U , the function $\omega(X): U \rightarrow \mathbb{R}$ is smooth on U .

We denote the real vector space of all smooth covector fields on M by $\mathfrak{X}^*M$ or $\Omega^1(M)$. For $f \in C^\infty(M)$, the covector field $f\omega$ is defined as

$$(f\omega)_p = f(p)\omega_p. \tag{11.5}$$

11.3 The Differential of a Function

Recall that the gradient of a smooth real-valued function f on $\mathbb{R}^n$ is defined as the vector field whose components are the partial derivatives of f

$$\text{grad } f = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}.$$

Although the partial derivatives of a smooth function cannot be interpreted in a coordinate-independent way as the components of a vector field, it turns out that they can be interpreted as the components of a covector field. This is the most important application of covector fields.

Definition 11.6. Let f be a smooth real-valued function on a smooth manifold M . We define a covector field df , called the **differential of f** , by

$$df_p(v) = v f \quad \text{for } v \in T_p M.$$

Proposition 11.7. The differential of a smooth function is a smooth covector field.

Proof. Clearly at each point $p \in M$, $df_p(v)$ depends linearly on v , so that df_p is indeed a covector at p . To see that df is smooth, we use Proposition 11.5 (3): for any smooth vector field X on M , the function $df(X)$ is smooth because it is equal to $X f$. $\square$

Let us compute df in coordinates or in other words, let's compute its coordinate representation. Let (x^i) be smooth coordinates on an open subset $U \subseteq M$, and let (λ^i) be the corresponding coordinate dual basis on U . Write df in coordinates as $df_p = A_i(p) \lambda^i|_p$ for some functions $A_i: U \rightarrow \mathbb{R}$; then the definition of df implies

$$A_i(p) = df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial}{\partial x^i} \Big|_p f = \frac{\partial f}{\partial x^i}(p).$$

This yields the following formula for the coordinate representation of df :

$$df_p = \frac{\partial f}{\partial x^i}(p) \lambda^i \Big|_p. \quad (11.6)$$

Thus, the component functions of df in any smooth coordinate chart are the partial derivatives of f with respect to those coordinates. As a result, the differential of f in coordinates is analogous to the gradient of f in local coordinates but df makes sense irrespective of any coordinate system.

Applying (11.6) to the special case in which f is one of the coordinate functions $x^j: U \rightarrow \mathbb{R}$, we obtain

$$dx^j|_p = \frac{\partial x^j}{\partial x^i}(p) \lambda^i|_p = \delta_i^j \lambda^i|_p = \lambda^j|_p.$$

In other words, the coordinate covector field λ^j is none other than the differential dx^j . Therefore, the formula (11.6) for df_p can be rewritten as

$$df_p = \frac{\partial f}{\partial x^i}(p) dx^i|_p,$$

or as an equation between covector fields instead of covectors:

$$df = \frac{\partial f}{\partial x^i} dx^i. \quad (11.11)$$

From now on, we abandon the notation λ^i for the coordinate coframe, and use dx^i instead.

Example 11.8. If $f(x, y) = x^2y \cos x$ on $\mathbb{R}^2$, then df is given by the formula

$$\begin{aligned} df &= \frac{\partial (x^2y \cos x)}{\partial x} dx + \frac{\partial (x^2y \cos x)}{\partial y} dy \\ &= (2xy \cos x - x^2y \sin x) dx + x^2 \cos x dy. \end{aligned}$$

Recall that for a smooth real-valued function $f: M \rightarrow \mathbb{R}$, we now have two different definitions for the differential of f at a point $p \in M$. Previously, we defined df_p as a linear map from $T_p M$ to $T_{f(p)} \mathbb{R}$. Now, we defined df_p as a covector at p , which is to say a linear map from $T_p M$ to $\mathbb{R}$. These are really the same object, once we take into account the canonical identification between $\mathbb{R}$ and $T_{f(p)} \mathbb{R}$; one easy way to see this is to note that both are represented in coordinates by the row matrix whose components are the partial derivatives of f .

Proposition 11.9 (Properties of the Differential). *Let M be a smooth manifold and let $f, g \in C^\infty(M)$.*

- (a) *If a and b are constants, then $d(af + bg) = a df + b dg$.*
- (b) *$d(fg) = f dg + g df$.*
- (c) *$d(f/g) = (g df - f dg)/g^2$ on the set where $g \neq 0$.*
- (d) *If f is constant, then $df = 0$.*

Proposition 11.10 (Functions with Vanishing Differentials). *If f is a smooth real-valued function on a smooth manifold M then $df = 0$ if and only if f is constant on each component of M .*

Proof. It suffices to assume that M is connected and show that $df = 0$ if and only if f is constant. One direction is immediate: if f is constant, then $df = 0$. Conversely, suppose $df = 0$, let $p \in M$, and let $\mathcal{C} = \{q \in M : f(q) = f(p)\}$. If q is any point in $\mathcal{C}$, let U be a smooth coordinate ball centred at q . From (11.11) we see that $\frac{\partial f}{\partial x^i} \equiv 0$ in U for each i , so by elementary calculus f is constant on U . This shows that $\mathcal{C}$ is open, and since it is closed by continuity, it must be all of M . Thus, f is everywhere equal to the constant $f(p)$. $\square$